

Docker & Kubernetes

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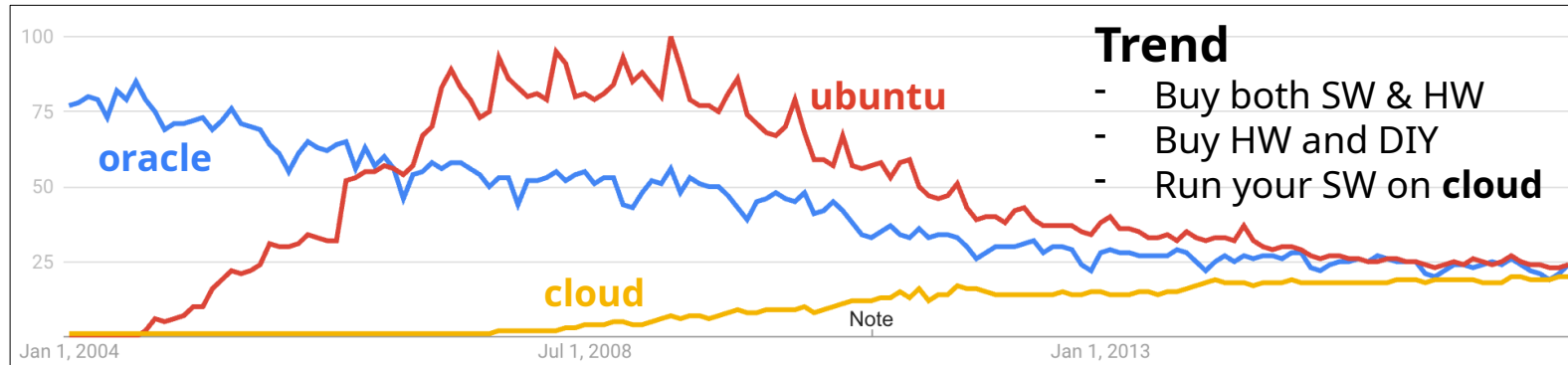
Big Data Tech. Lab in SK telecom

- Discovery Group
- Predictive Maintenance Group
- Manufacturing Solution Group
 - Groups making own solutions
- **Technology and Architecture Leading Group**
 - Big data processing engine
 - Advanced analytics algorithms
 - **Systematize service deployment and service operation on cluster**
 - **Docker**
 - **Kubernetes**

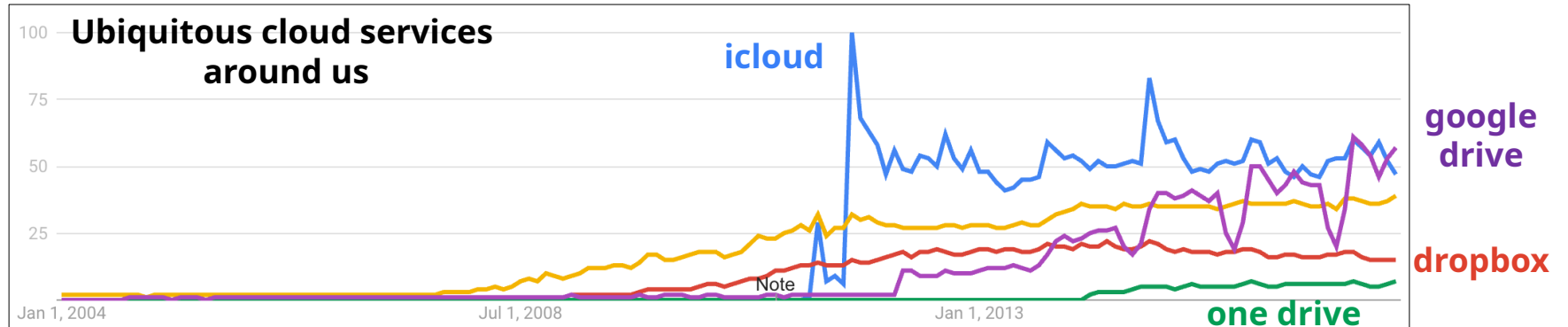
Prepare for an era of **cloud** with **Docker** and **Kubernetes**

* technology trend in USA (2004-2017)

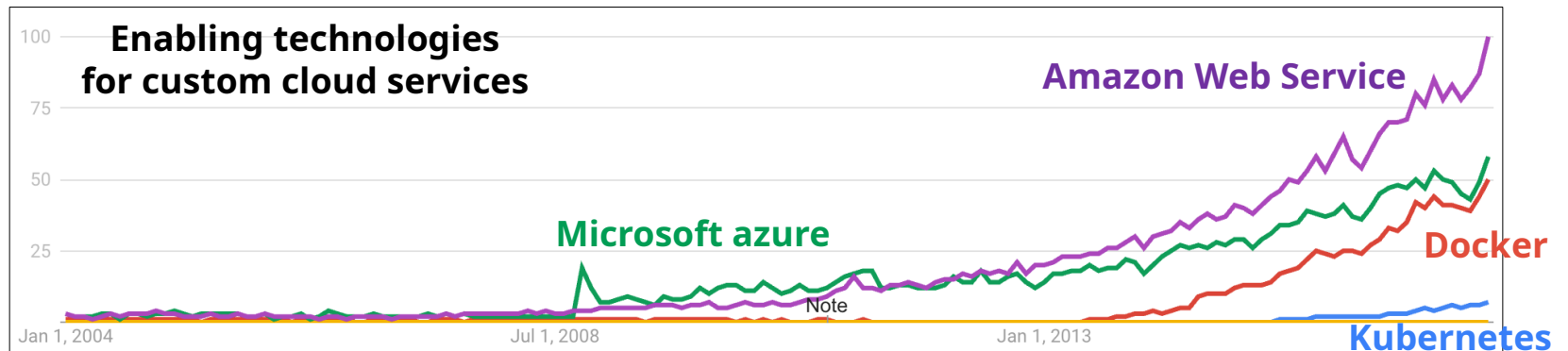
Major technologies



Cloud services for users

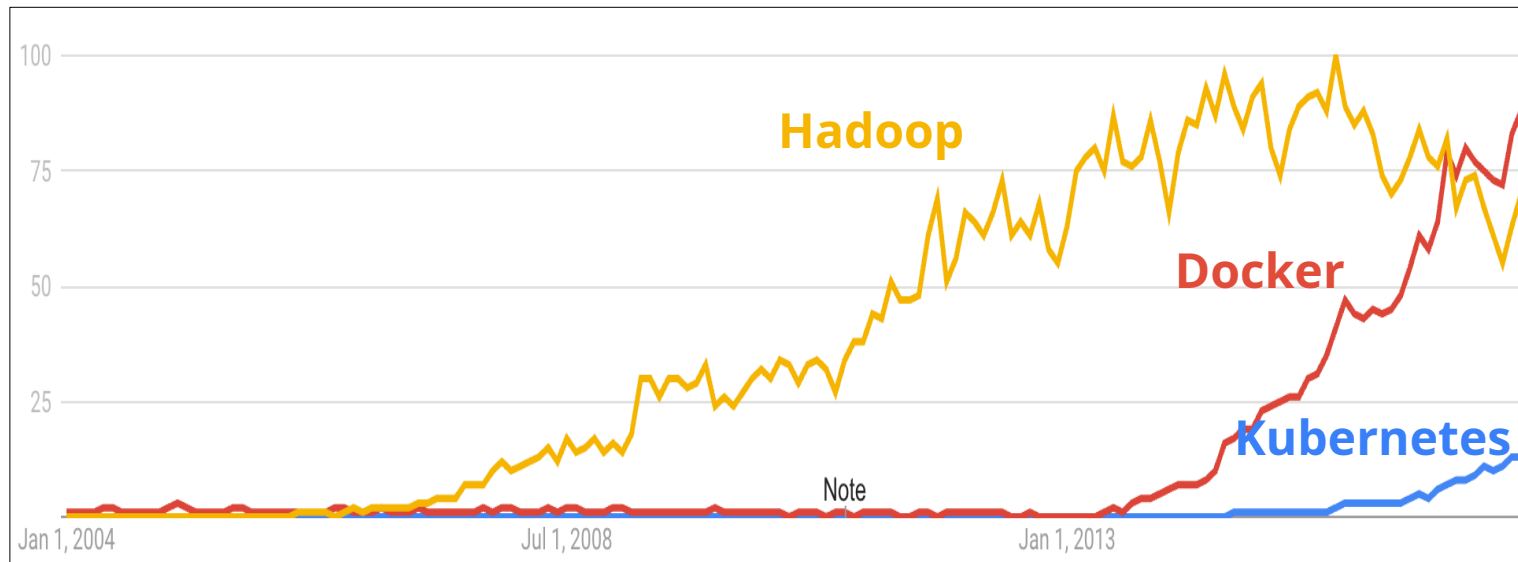


Cloud technologies for service providers

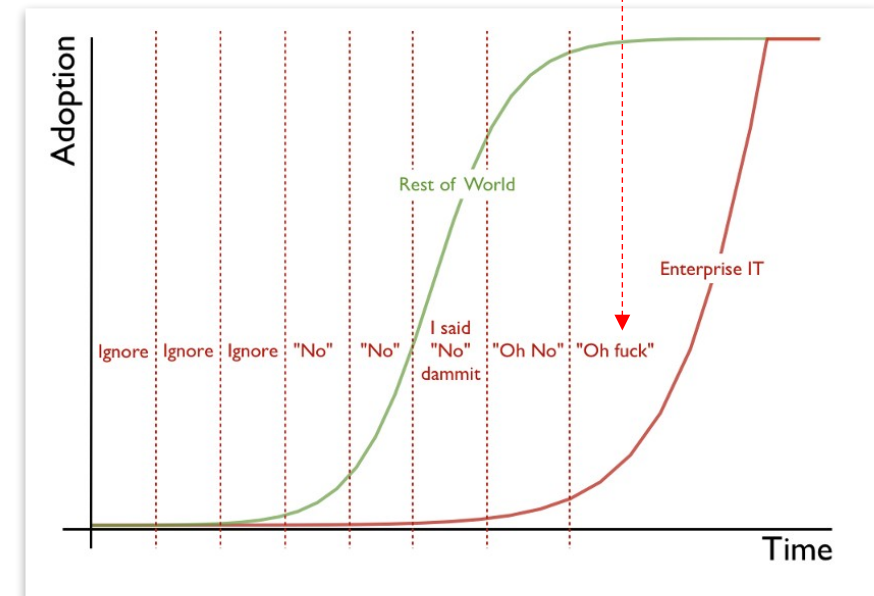


Overview & Conclusion

- **Docker** to build ***portable*** software
 - Build your software upon **Docker**
 - Then distribute it anywhere (even on MS Azure and Amazon Web Service)
- **Kubernetes** to orchestrate multiple **Docker** instances
- Start using **Docker** and **Kubernetes** before too late!
 - Google has been using container technologies more than 10 years



Popularity of **Docker** and **Kubernetes**



The Enterprise IT Adoption Cycle

Docker

Motivation

Enabling technologies for Docker

How to use Docker

Docker came to save us from the **dependency hell**

Docker

Dependency hell



Portable software

Dependency hell

Development
environment

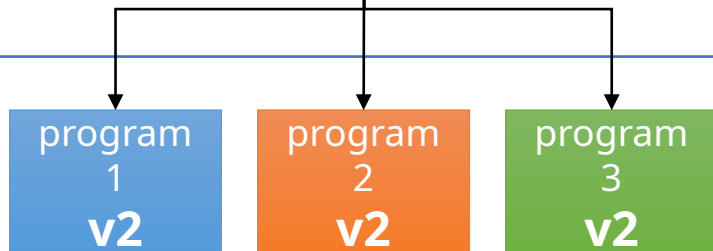


Production
environment



Your program

depends on



Package manager

Your program

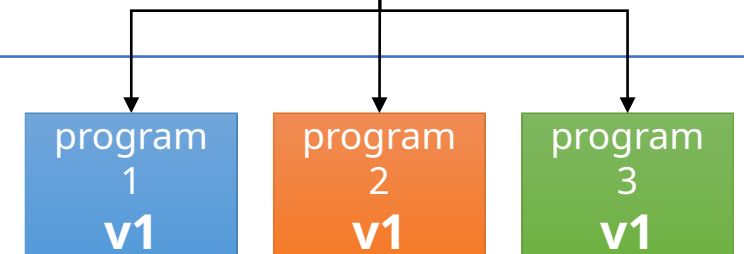
depends on



Package manager

Customer program

depends on



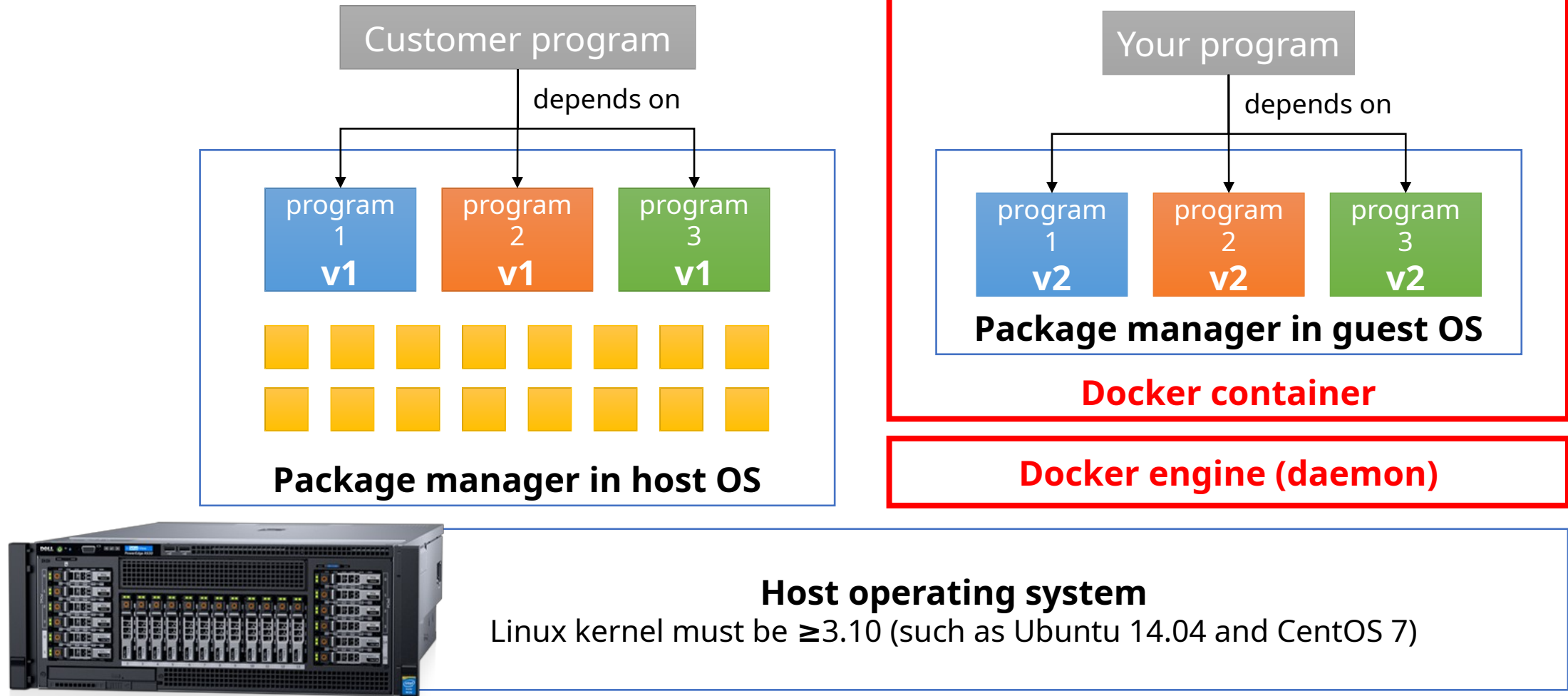
conflict!

Few choices left to you



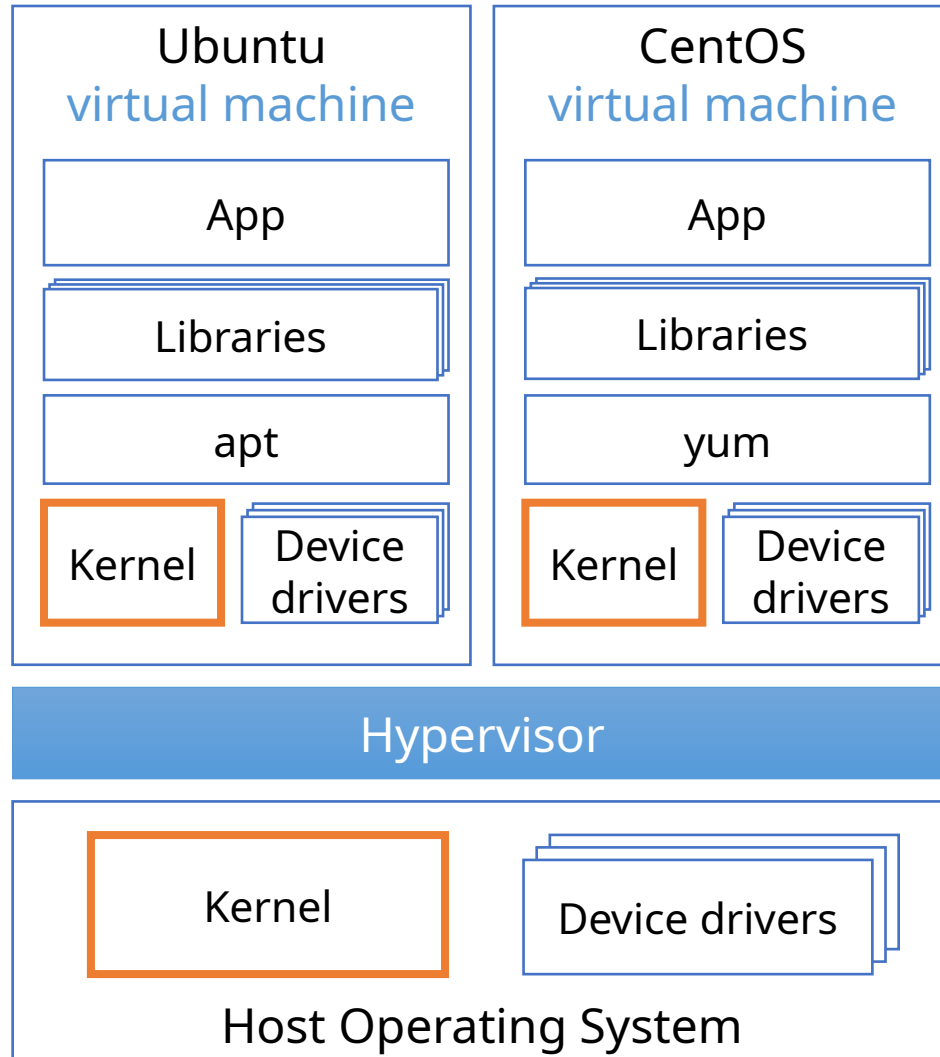
1. Convince your customer (a.k.a. 甲方)
2. Install all the dependencies manually (without the package manager)
3. Modify your program to make it depend v1

Use **Docker** for isolating your application

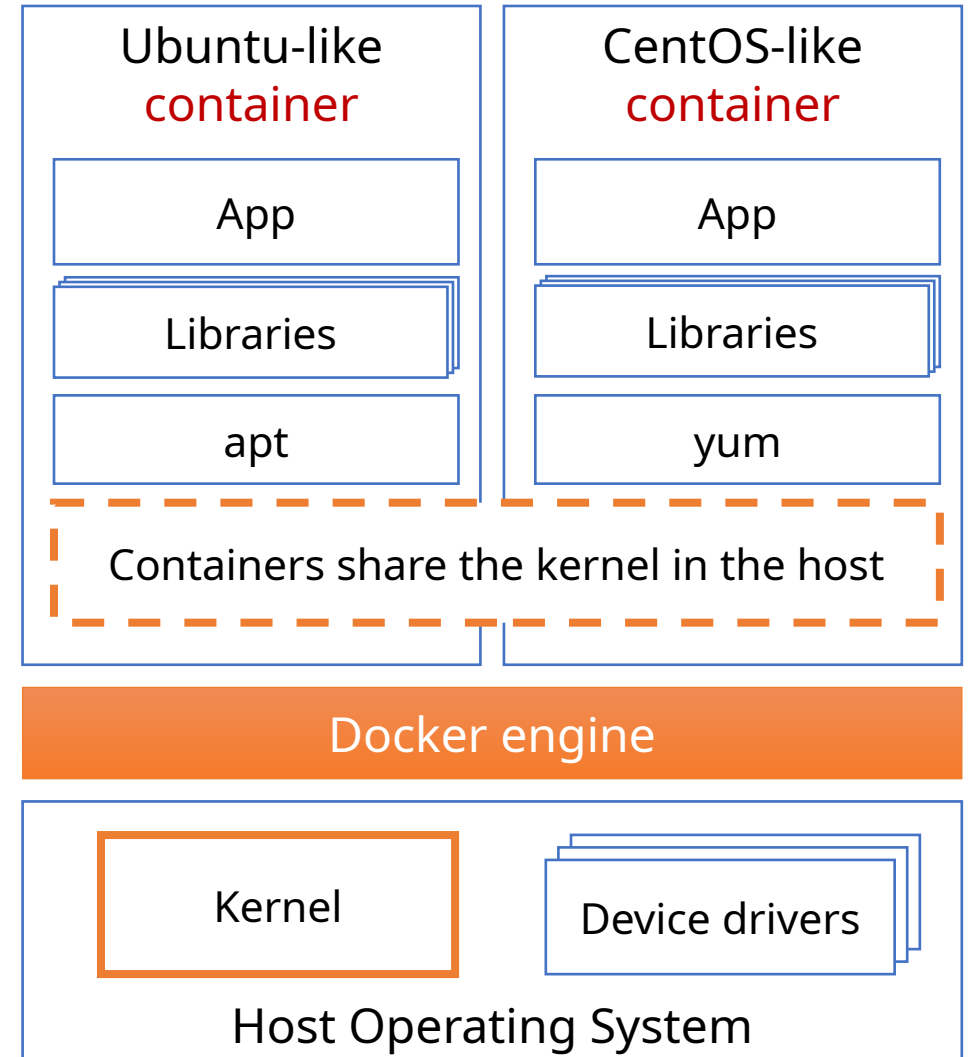


Virtual machines and docker containers

Virtual machines

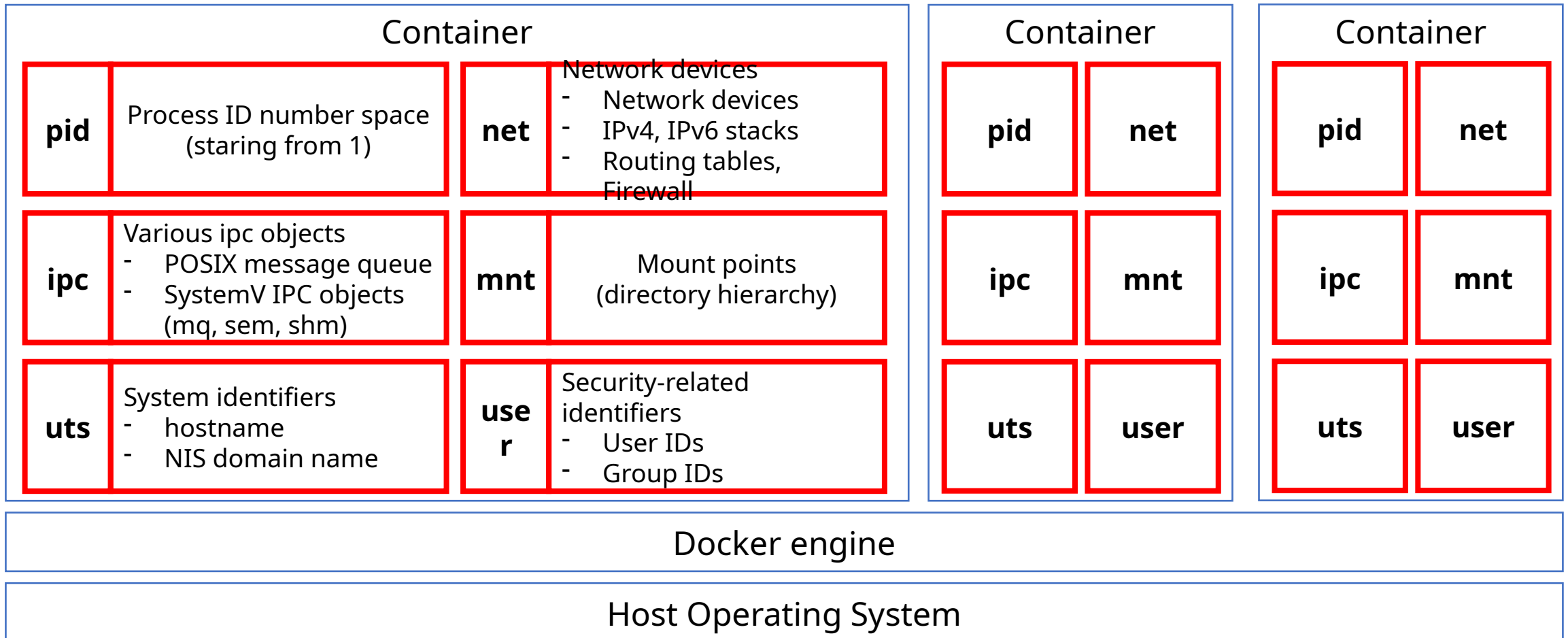


Docker containers



Linux namespaces – what makes **isolated** environments in a host OS

Six namespaces are enough to give *an illusion of running inside a virtual machine*



Analogy between program and docker

Program

Source code

```
class Square {
    public static void main(String[] args){
        if (args.length != 1) {
            System.exit(1);
        }
        int number = Integer.parseInt(args[0]);
        int square = number * number;
        System.out.println(square);
    }
}
```

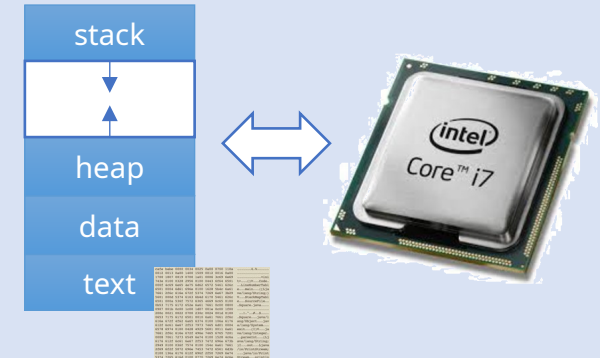
Byte/machine code (read only)

```
cafe babe 0000 0034 0025 0a00 0700 110a .....4.%.....
0012 0013 0a00 1400 1509 0012 0016 0a00 .....
1700 1807 0019 0700 1a01 0006 3c69 6e69 .....<ini
743e 0100 0328 2956 0100 0443 6f64 6501 t>...()V...Code.
000f 4c69 6e65 4e75 6d62 5572 5461 626c ..LineNumberTabl
6501 0004 6461 696e 0100 1628 5b4c 6a61 e...main...([Lja
7661 2f6c 616e 672f 5374 7269 6e67 3b29 va/lang/String;)
5601 000d 5374 6163 6b4d 6170 5461 626c V...StackMapTabl
6501 000a 536f 7572 6365 4669 6c65 0100 e...SourceFile..
0b53 7175 6172 652e 6a61 7661 0c00 0800 .Square.java....
0907 001b 0c00 1c00 1d07 001e 0c00 1f00 .....
200c 0021 0022 0700 230c 0024 001d 0100 ..!..".$.
0653 7175 6172 6501 0010 6a61 7661 2f6c .Square...java/l
616e 672f 4f62 6a65 6374 0100 106a 6176 ang/Object...jav
612f 6c61 6e67 2f53 7973 7465 6d01 0004 a/lang/System...
6578 6974 0100 0428 4929 5601 0011 6a61 exit...()V...ja
7661 2f6c 616e 672f 496e 7465 6765 7201 va/lang/Integer.
0008 7061 7273 6549 6e74 0100 1528 4c6a ..parseInt...([j
6176 612f 6c61 6e67 2f53 7472 696e 673b ava/lang/String;
2949 0100 036f 7574 0100 154c 6a61 7661 )I...out...Ljava
2f69 6f2f 5072 696e 7453 7472 6561 6d3b /io/PrintStream;
0100 136a 6176 612f 696f 2f50 7269 6e74 ...java/io/Print
5374 7265 616d 0100 0770 7269 6e74 6c6e Stream...println
```

compile
➡

execute
➡

Process (read only)



Dockerfile

```
FROM scratch
ADD ubuntu-trusty-core-cloudimg-amd64-root.tar.gz /

# a few minor docker-specific tweaks
# see https://github.com/docker/docker/blob/9a9fc81af8fb5d98b8e0c8748716226fadb3735c/contrib/mkimage/debootstrap
RUN set -xe \
    \
    # https://github.com/docker/docker/blob/9a9fc81af8fb5d98b8e0c8748716226fadb3735c/contrib/mkimage/debootstrap#L40-L48
    && echo '#!/bin/sh' > /usr/sbin/policy-rc.d \
    && echo 'exit 101' >> /usr/sbin/policy-rc.d \
    && chmod +x /usr/sbin/policy-rc.d \
    \
    # https://github.com/docker/docker/blob/9a9fc81af8fb5d98b8e0c8748716226fadb3735c/contrib/mkimage/debootstrap#L54-L56
    && dpkg-divert --local --rename --add /sbin/initctl \
    \
    # make systemd-detect-virt return "docker"
    # See: https://github.com/systemd/systemd/blob/aa0c34279ee40bce2f9681b496922dedbadfca19/src/basic/virt.c#L434
    RUN mkdir -p /run/systemd && echo 'docker' > /run/systemd/container

# overwrite this with 'CMD []' in a dependent Dockerfile
CMD ["bin/bash"]
```

Docker

build
➡

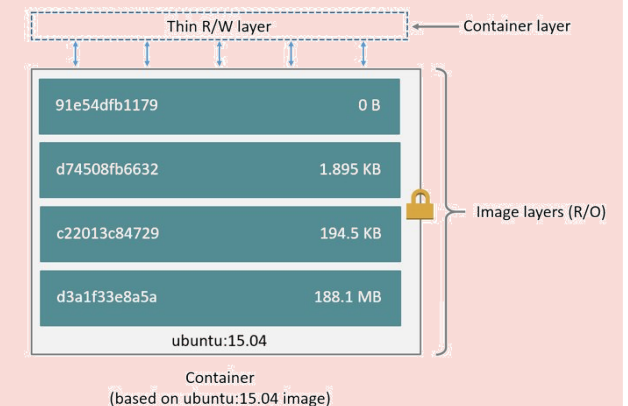
Docker image (read-only **layers**)

91e54dfb1179	0 B
d74508fb6632	1.895 KB
c22013c84729	194.5 KB
d3a1f33e8a5a	188.1 MB
ubuntu:15.04	

Image

run
➡

Docker container (read-only **layers** + **writable layer**)



How to define **an image** and run **a container** from it?

1) Write Dockerfile

- Specify to install **python** with **pip** on **ubuntu**
- Tell **pip** to install **numpy**

```
FROM ubuntu

RUN apt-get update \
    && apt-get -y install python-dev python-pip \
    && rm -rf /var/lib/apt/lists/*

RUN pip install numpy

CMD ["python"]
```

2) Build **an image** from Dockerfile

- Execute each line of Dockerfile to build **an image**

```
~/tmp/docker/numpy$ docker build -t numpy .
Sending build context to Docker daemon 3.072 kB
Step 1/4 : FROM ubuntu
----> 0ef2e08ed3fa
Step 2/4 : RUN apt-get update
    && apt-get -y install python-dev python-pip
    && rm -rf /var/lib/apt/lists/*
----> a6586eb5b798
Step 3/4 : RUN pip install numpy
----> 7e1ae658614
Step 4/4 : CMD python
----> dcc7c9deb606
Successfully built dcc7c9deb606
```

3) Execute **a Docker container** from **the image**

```
~/tmp/docker/numpy$ docker run --tty --interactive numpy
Python 2.7.12 (default, Nov 19 2016, 06:48:10)
[GCC 5.4.0 20160609] on linux2
Type "help", "copyright", "credits" or "license" for more information.
>>> import numpy as np
import numpy as np
>>> np.array([1,2,3,4]) * 100
np.array([1,2,3,4]) * 100
array([100, 200, 300, 400])
>>>
```

1 to N relationship between **image** and **container**

```
~/tmp/docker/numpy$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
numpy	latest	dcc7c9deb606	15 minutes ago	489 MB

```
~/tmp/docker/numpy$ docker run --tty --interactive numpy
~/tmp/docker/numpy$ docker run --tty --interactive numpy
~/tmp/docker/numpy$ docker run --tty --interactive numpy
~/tmp/docker/numpy$ docker run --tty --interactive numpy
~/tmp/docker/numpy$ docker run --tty --interactive numpy
~/tmp/docker/numpy$ docker ps -a
```

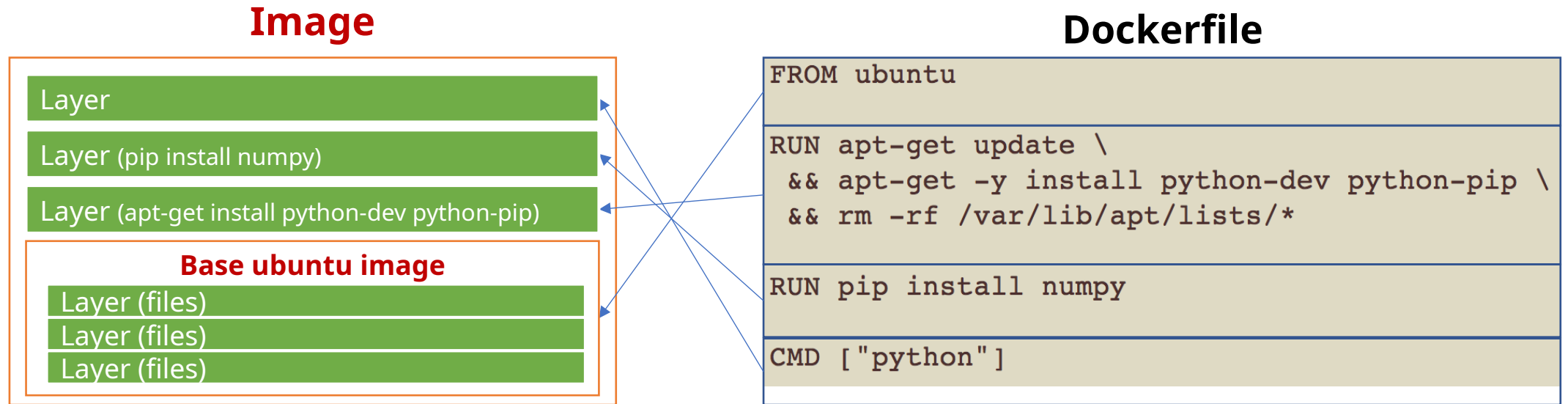
CONTAINER ID	IMAGE	COMMAND
d99c0def8f2b	numpy	"python"
9db9f0226e14	numpy	"python"
e4bbf42cefa9	numpy	"python"
f8ca1be0d682	numpy	"python"
fb439aa3d49a	numpy	"python"

Execute **five containers** from **an image**

Q) Five containers take up 2,445MB (=489MB*5) in the host?

A) No due to **image layering & sharing**

Images consists of **layers** each of which is a **set of files**



- **Instructions** (FROM, RUN, CMD, etc) create **layers**
 - **Base images** (imported by "FROM") also consist of layers
- If a file exists in multiple layers, the one in the upper layer is seen

Docker container

- **A container is just a thin read/write layer**
 - base images are not copied to containers
- Copy-On-Write (**COW**)
 - When **a file in the base image** is modified,
 - **copy** the file to the R/W layer
 - and **then modify** the copied file

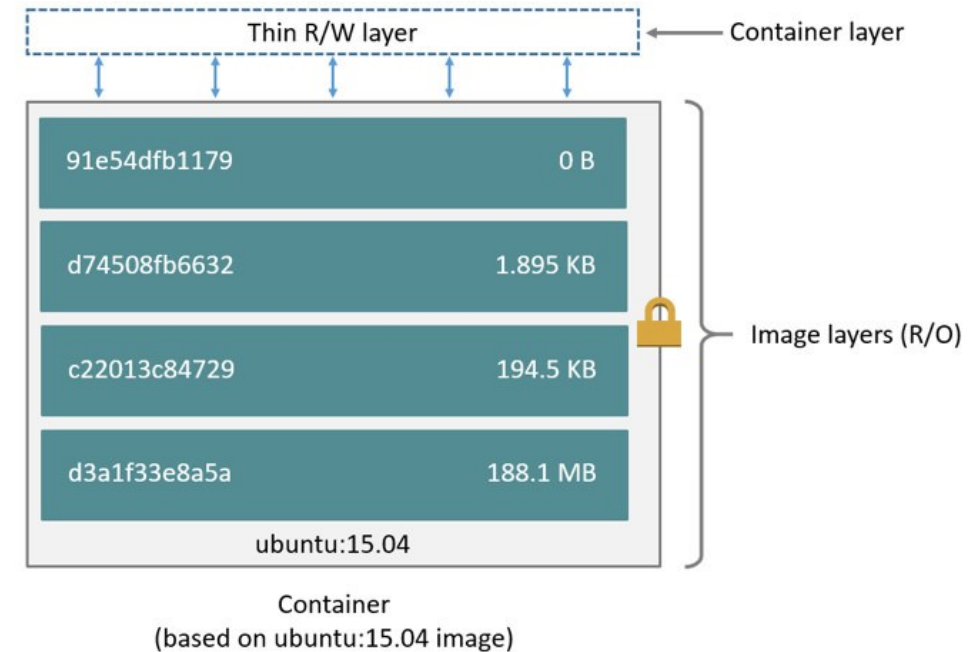
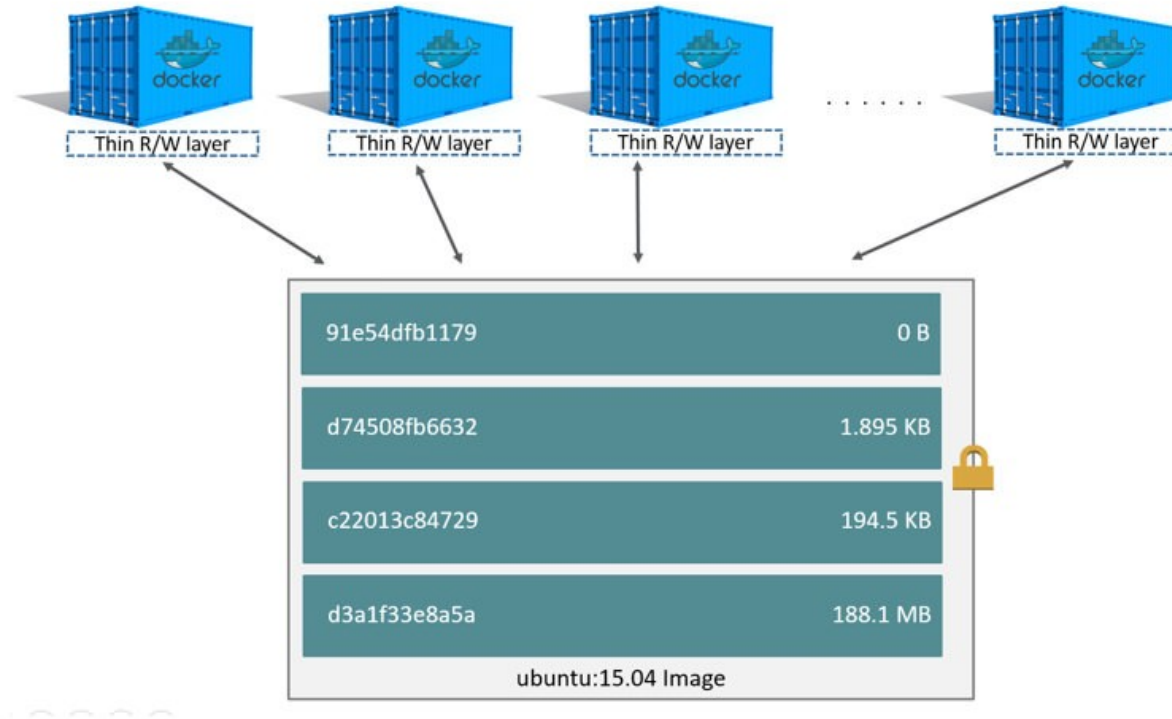


Image sharing between containers



ubuntu:15.04 image (~188MB) does not copied to all containers

Layer sharing between images

If multiple Dockerfiles

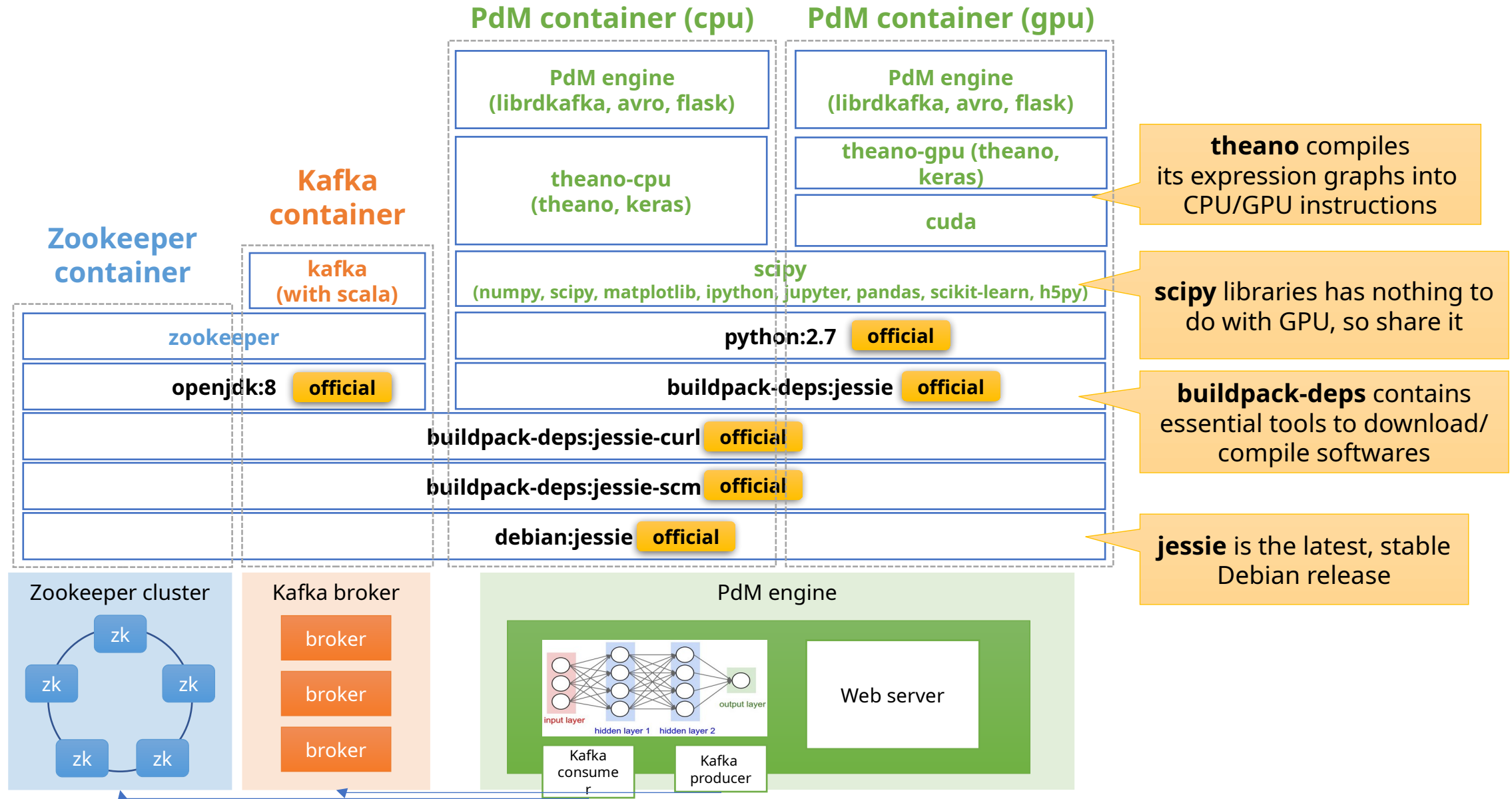
1. start from the same base image
2. share a sequence of instructions (one RUN instruction in a below example)

FROM ubuntu		FROM ubuntu
RUN apt-get update \ && apt-get -y install python-dev python-pip \ && rm -rf /var/lib/apt/lists/*		RUN apt-get update \ && apt-get -y install python-dev python-pip \ && rm -rf /var/lib/apt/lists/*
RUN pip install numpy		RUN pip install matplotlib
CMD ["python"]	numpy Dockerfile	CMD ["python"] matplotlib Dockerfile

, then docker engine automatically reuses existing layers

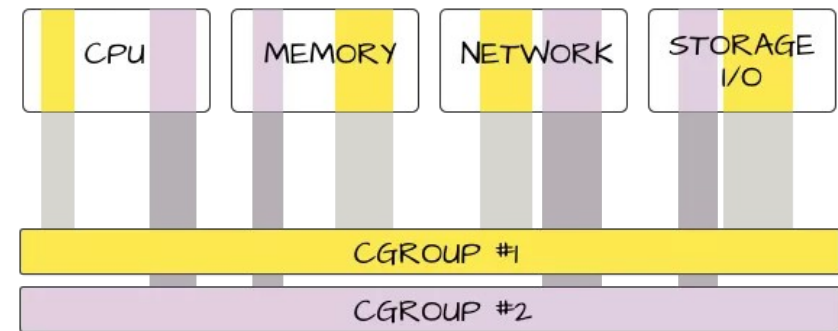
~/tmp/docker/matplotlib\$ docker history numpy				~/tmp/docker/matplotlib\$ docker history matplotlib			
IMAGE	CREATED	CREATED BY	SIZE	IMAGE	CREATED	CREATED BY	SIZE
dcc7c9deb606	22 hours ago	/bin/sh -c #(nop) CMD ["python"]	0 B	66106d688bc9	20 seconds ago	/bin/sh -c #(nop) CMD ["python"]	0 B
7eela658614	22 hours ago	/bin/sh -c pip install numpy	92.1 MB	21d73293ac73	21 seconds ago	/bin/sh -c pip install matplotlib	150 MB
a6586eb5b798	22 hours ago	/bin/sh -c apt-get update && apt-get -y i...	267 MB	a6586eb5b798	22 hours ago	/bin/sh -c apt-get update && apt-get -y i...	267 MB
0ef2e08ed3fa	35 hours ago	/bin/sh -c #(nop) CMD ["/bin/bash"]	0 B	0ef2e08ed3fa	35 hours ago	/bin/sh -c #(nop) CMD ["/bin/bash"]	0 B
<missing>	35 hours ago	/bin/sh -c mkdir -p /run/systemd && echo '...	7 B	<missing>	35 hours ago	/bin/sh -c mkdir -p /run/systemd && echo '...	7 B
<missing>	35 hours ago	/bin/sh -c sed -i 's/^#\s*(deb.*universe\...	1.9 kB	<missing>	35 hours ago	/bin/sh -c sed -i 's/^#\s*(deb.*universe\...	1.9 kB
<missing>	35 hours ago	/bin/sh -c rm -rf /var/lib/apt/lists/*	0 B	<missing>	35 hours ago	/bin/sh -c rm -rf /var/lib/apt/lists/*	0 B
<missing>	35 hours ago	/bin/sh -c set -xe && echo '#!/bin/sh' >...	745 B	<missing>	35 hours ago	/bin/sh -c set -xe && echo '#!/bin/sh' >...	745 B
<missing>	35 hours ago	/bin/sh -c #(nop) ADD file:efb254bc677d66d...	130 MB	<missing>	35 hours ago	/bin/sh -c #(nop) ADD file:efb254bc677d66d...	130 MB

Example of stacking docker images



Enabling technologies for docker (wrap-up)

- **Linux namespaces** (covered)
 - To isolate system resources
 - pid, net, ipc, mnt, uts, user
 - It makes a secure & isolate environment (like a VM)
- **Advanced multi-layer unification File System** (covered)
 - Image layering & sharing
- **Linux control groups** (not covered)
 - To track, limit, and isolate resources
 - CPU, memory, network, and IO



* <https://mairin.wordpress.com/2011/05/13/ideas-for-a-cgroups-ui/>

Docker topics not covered here

- How to install **Docker engine**
- What are the docker instructions other than **FROM**, **RUN**, and **CMD**
 - ENV / ADD / ENTRYPOINT / LABEL / EXPOSE / COPY / VOLUME / WORKDIR / ONBUILD
- How to push **local Docker images** to docker hub
- How to pull **remote images** from docker hub
- ...

Consult with <https://docs.docker.com/engine/getstarted/>

Kubernetes

Motivation

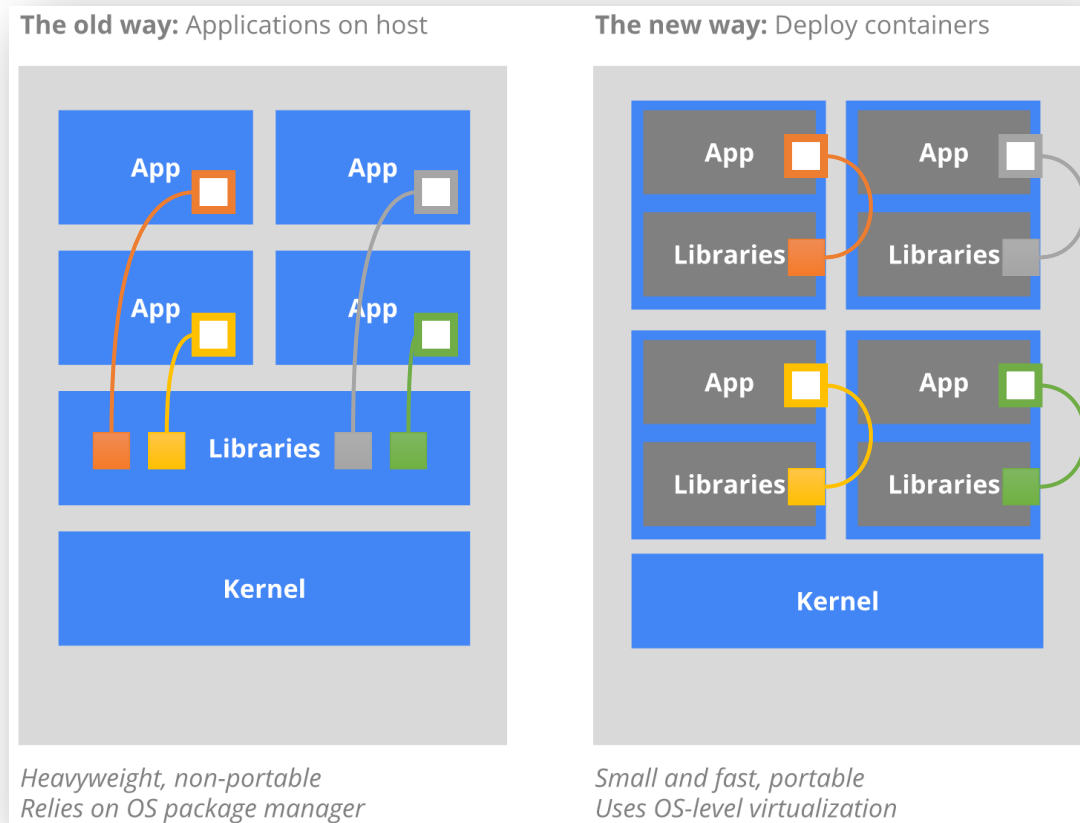
A motivating example

Disclaimer

- The purpose of this section is to briefly explain Kubernetes **without details**
- For a detailed explanation with the exact Kubernetes terminology, see the following slide
 - <https://www.slideshare.net/ssuser6bb12d/kubernetes-introduction-71846110>

What is Kubernetes for?

Container-based virtualization + **Container orchestration**
To satisfy common needs in **production**



replicating application instances
naming and discovery
load balancing
horizontal auto-scaling
co-locating helper processes
mounting storage systems
distributing secrets
application health checking
rolling updates
resource monitoring
log access and ingestion
...

from the official site : <https://kubernetes.io/docs/whatisk8s/>

Why **Docker** with **Kubernetes**?

- A mission of our group
 - **Systematize** **service deployment** and **service operation** on cluster
 - I believe that **systematizing smth.** is **to minimize human efforts on smth.**
- How to **minimize human efforts** on **service deployment**?
 - Make software portable using **a container technology**
 - **Docker** (chosen for its maturity and popularity)
 - **Rkt** from CoreOS (alternative)
 - Build images and run containers anywhere
 - Your laptop, servers, on-premise clusters, even cloud
- How to **minimize human efforts** on **service operation**?
 - Inform **a container orchestration runtime** of **service specification**
 - **Kubernetes** from Google (chosen for its **maturity** and **expressivity**)
 - **Docker swarm** from Docker
 - Define **your specification** and then the runtime operates your services as you wish

Kubernetes architecture



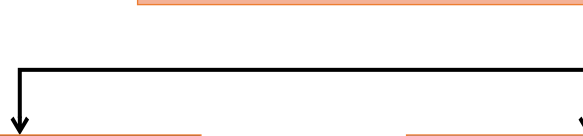
Service specification (written in yaml)

- Execute a web-server image
- Two replicas for LB & HA
 - 3GB memory each



Server

- **REST API server** with a K/V store
- **Scheduler**
 - Find suitable machines for containers
- **Controller manager**
 - **Current** state $\neq$ **Desired** state
 - Make changes if states go **undesirable**

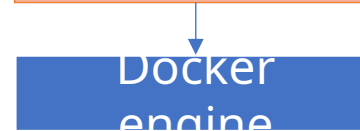


Node agent



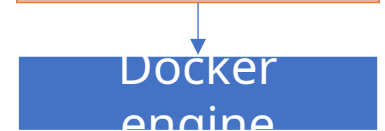
container
(3GB)

Node agent



container
(3GB)

Node agent



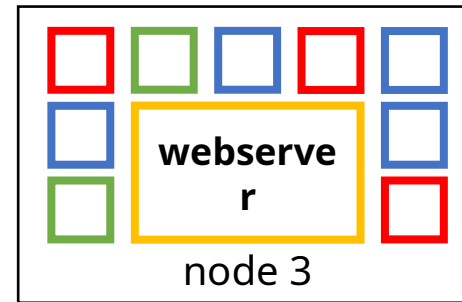
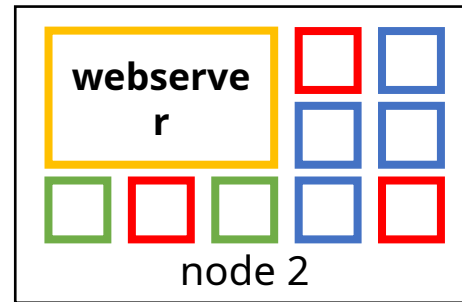
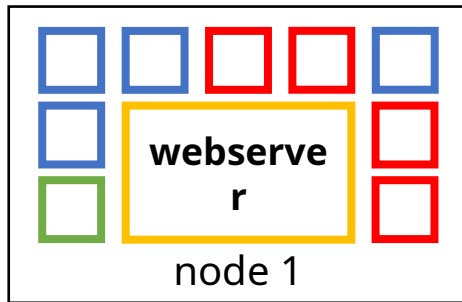
container
(3GB)

*Ensure a specified
of replicas running
all the time*

Web server example

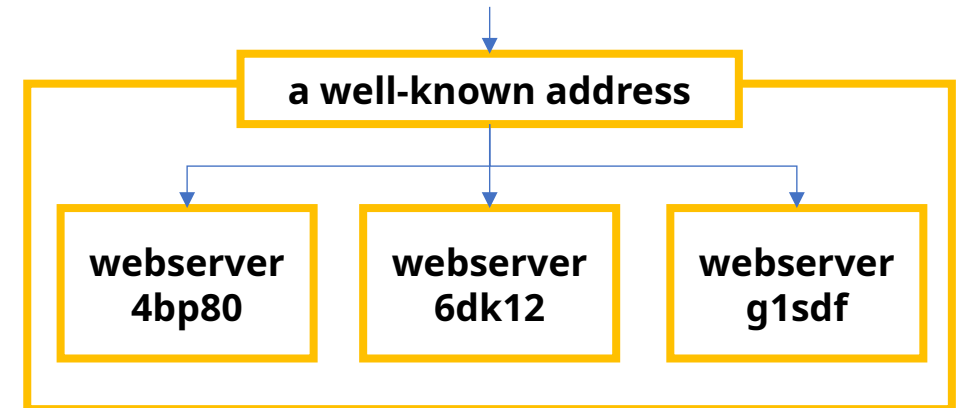


Want to launch 3 replicas
for **high availability** and **load balancing**



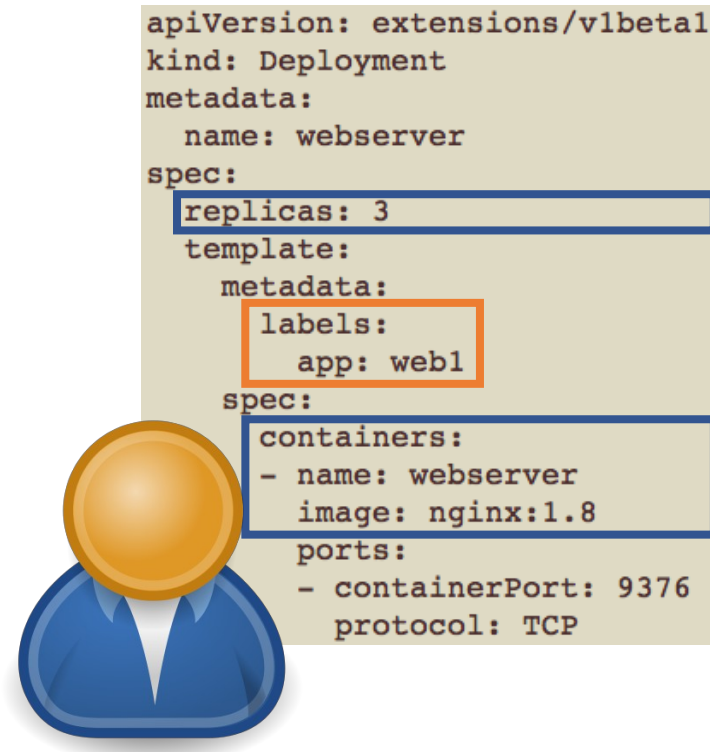
How to achieve the followings?

- Users must be unaware of the replicas
- Traffic is evenly distributed to replicas

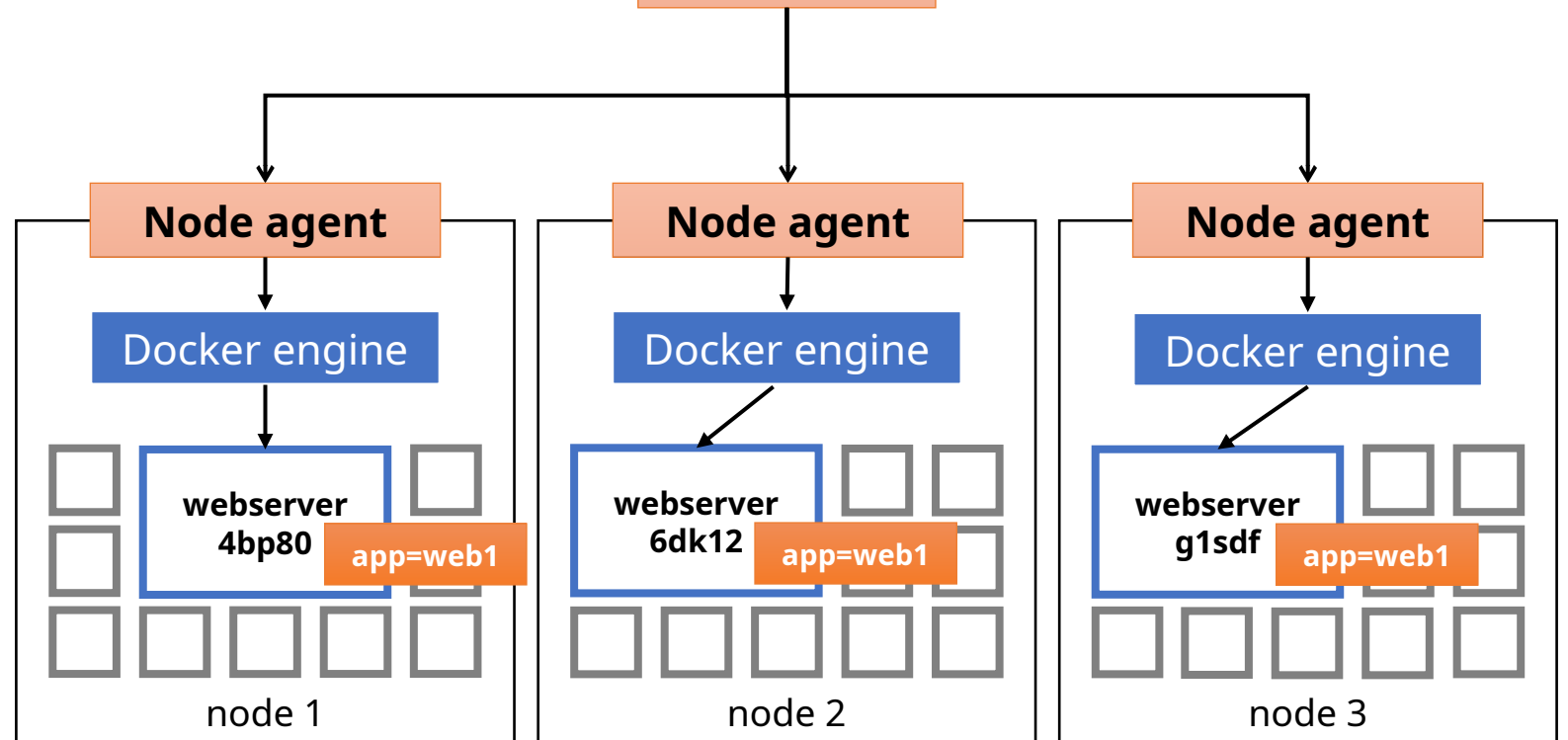


It's a piece of cake with Kubernetes!

How to replicate your service instances

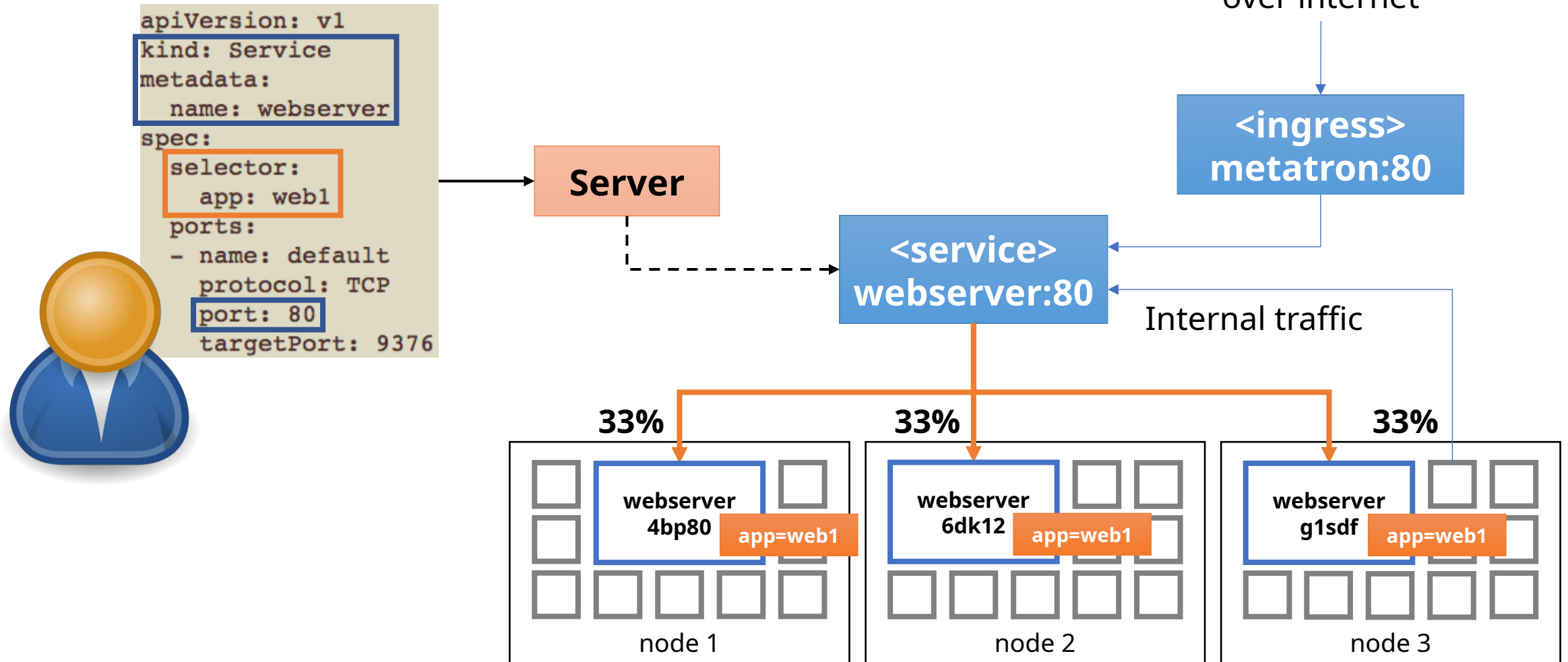


Specify **your Docker image** and a **replication factor** using **Deployment Server**



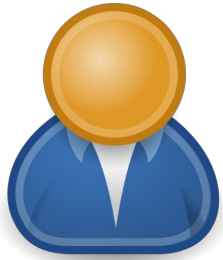
Specify a **common label** to group containers with different names

Define a **service** to do round-robin forwarding



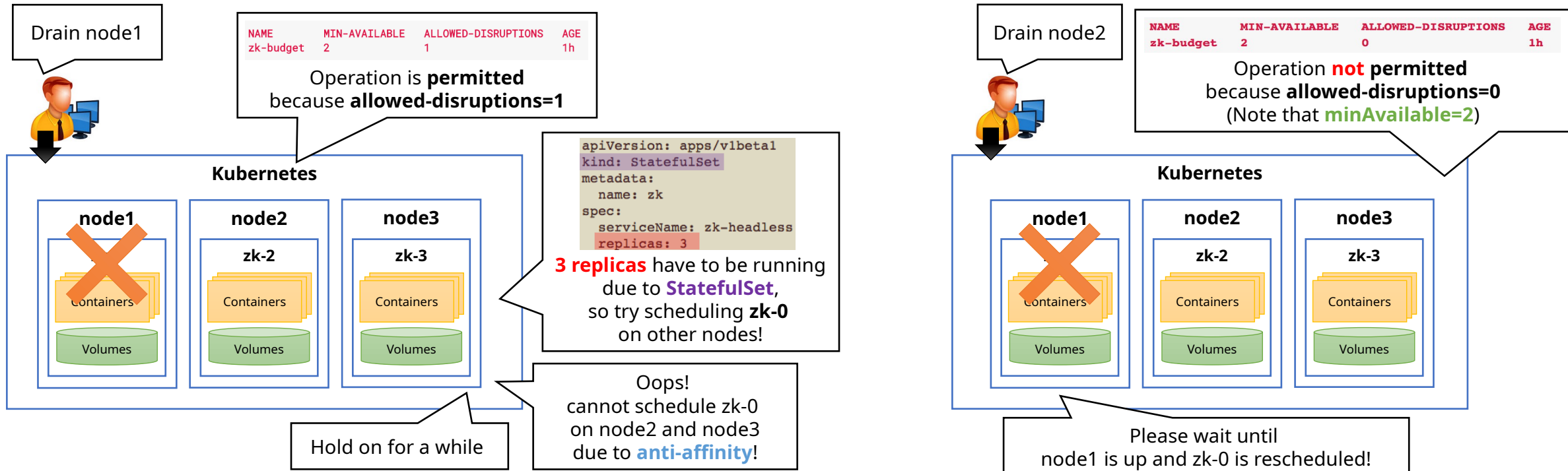
*Kubernetes runs **its own DNS server** for name resolution
Kubernetes **manipulates iptables** on each node **to proxy traffic***

How to guarantee a certain # of running containers during maintenance

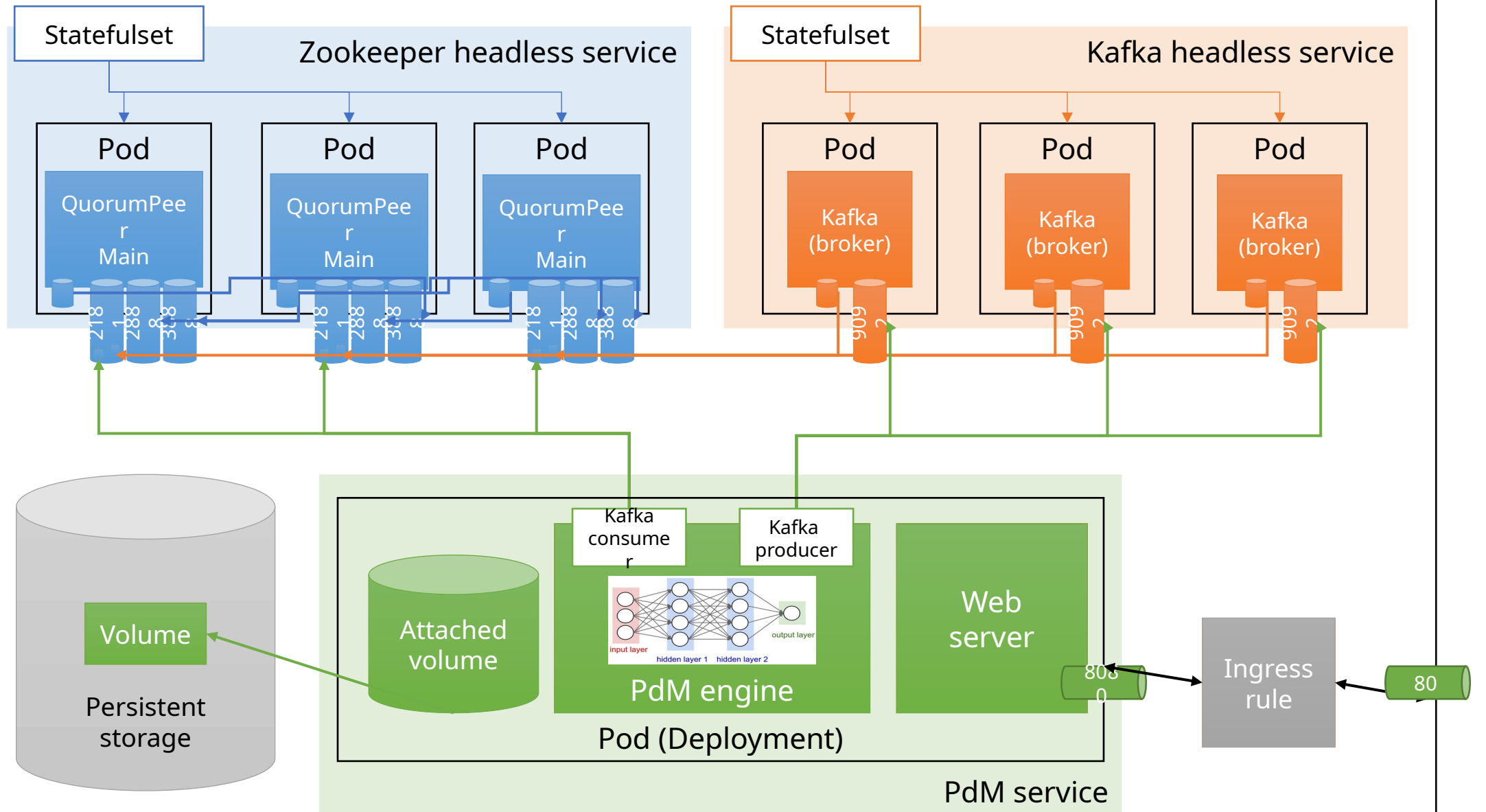


```
apiVersion: policy/v1beta1
kind: PodDisruptionBudget
metadata:
  name: zk-budget
spec:
  selector:
    matchLabels:
      app: zk
  minAvailable: 2
```

Define **disruption budget** to specify requirement for the **minimum available containers**

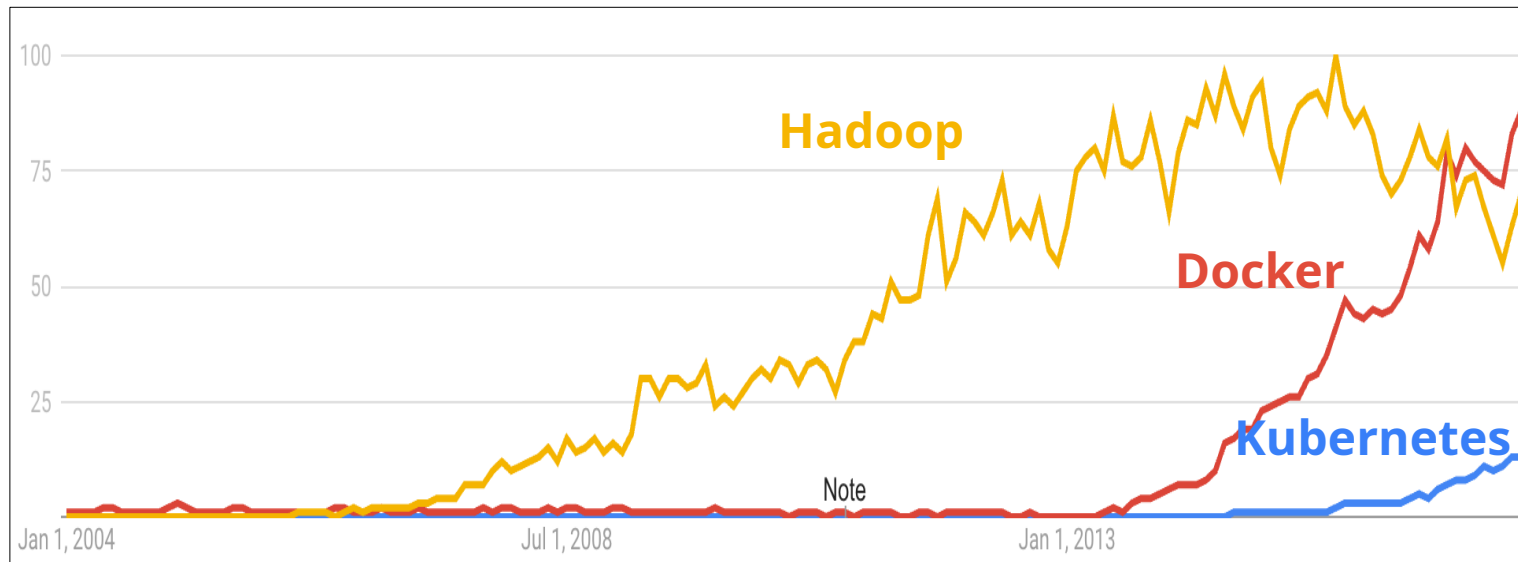


PdM Kubernetes cluster

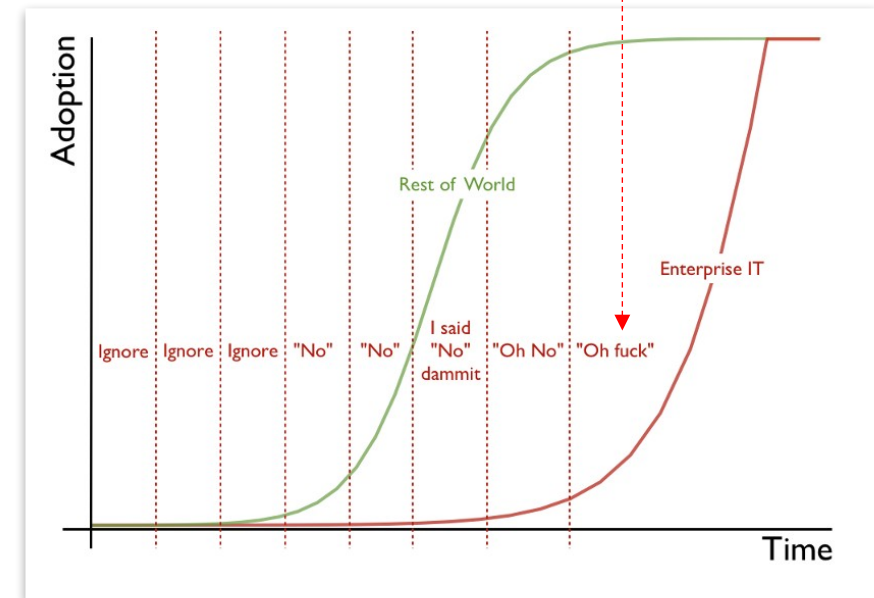


Overview & Conclusion

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 - Build your software upon Docker
 - Then distribute it anywhere (even on MS Azure and AWS)
- **Kubernetes** to orchestrate multiple **Docker** instances
- Start using **Docker** and **Kubernetes** before too late!
 - Google has been using container technologies more than 10 years



Popularity of **Docker** and **Kubernetes**



The Enterprise IT Adoption Cycle

the end